INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, KURNOOL

Department of Electronics and Communication Engineering

Electronic Prototyping and Manufacturing (EPM) Laboratory

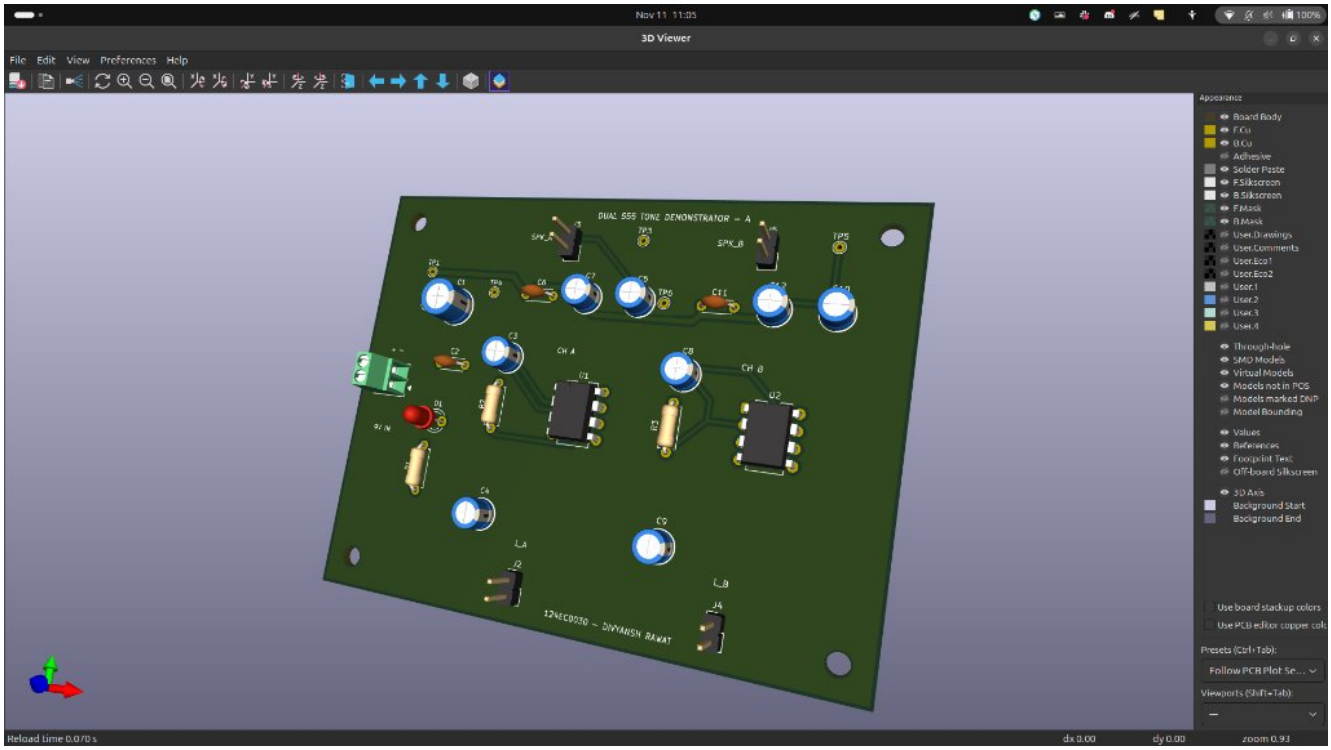
End Semester Practical Examination — Laboratory Report

Student Name:	Divyansh Rawat	**Roll Number:**	124EC0030
Course Faculty:	Dr. Eswaramoorthy K V	**Date of Exam:**	10-11-2026
Project Title:	Dual Inductance-Sensitive Tone Demonstrator - A	**Exam Pattern:**	Layout Variant A (90 x 70 mm)
Design Environment:	KiCad 9.0.8 (x86_64 Linux)	**DRC / ERC Status:**	**0 Errors / 0 Violations**

EXAM QUESTION SPECIFICATIONS (VARIANT A):

Implement two copies of the source 555 inductance-sensitive oscillator. Use an external inductor connector and speaker connector for each channel. Supply: 9 V DC. Use a 90 x 70 mm rectangular board. Place the supply connector on the left edge, input/user controls along the lower edge and outputs along the upper edge. Keep inductive/timing loops short; label L A/B and SPK A/B connector polarity where applicable. Mandatory circuit clarifications: No search-coil winding, detection-distance target or field calibration is required. This is a layout-only oscillator demonstrator, not a validated general metal detector. Use exact capacitor connections from the schematic. Assign verified through-hole footprints and complete a two-copper-layer PCB using both F.Cu and B.Cu.

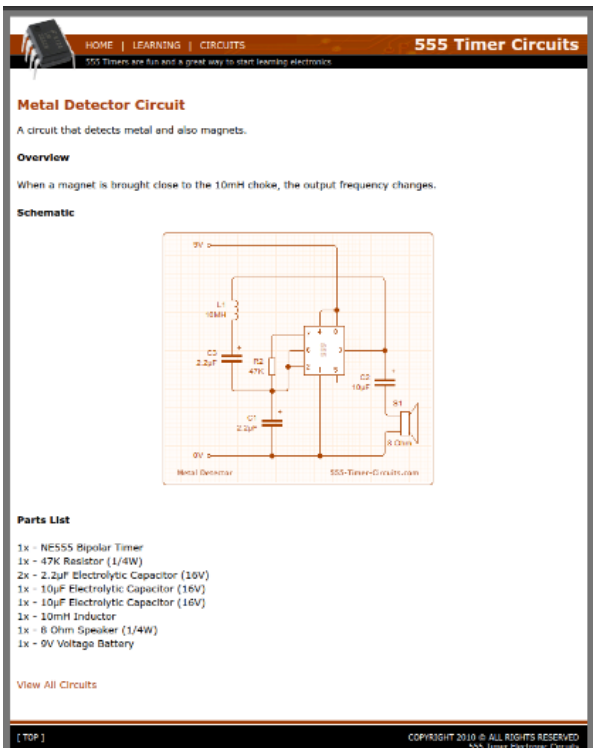
3D BOARD PREVIEW (REALISTIC GREEN SOLDERMASK):



1. CIRCUIT DIAGRAM (REFERENCE CIRCUIT SPECIFICATION)

Source Reference: **555-Timer-Circuits.com Metal Detector / Dual Inductance-Sensitive Tone Generator**. The circuit operates as an astable multivibrator where oscillation frequency is modulated by the inductive reactance of choke L1 in conjunction with timing capacitors and feedback resistor R2. When a metallic object is brought into proximity with the external choke, the inductance shifts, producing an audible frequency change.

Source Website & Parts List:



HOME | LEARNING | CIRCUITS

555 Timer Circuits

555 Timers are fun and a great way to start learning electronics

Metal Detector Circuit

A circuit that detects metal and also magnets.

Overview

When a magnet is brought close to the 10mH choke, the output frequency changes.

Schematic

Parts List

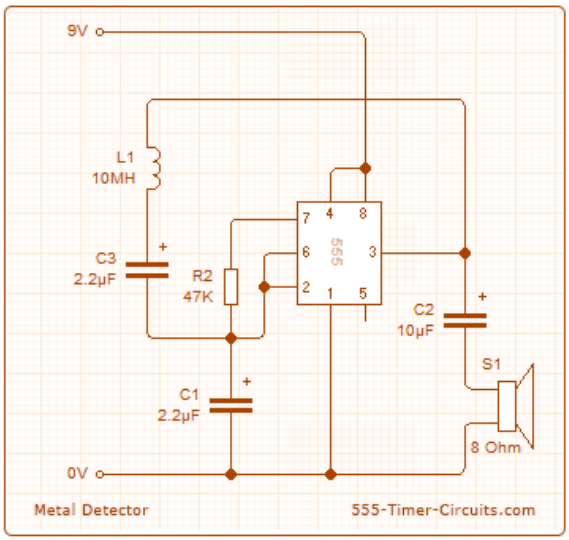
1x - NE555 Bipolar Timer
1x - 47K Resistor (1/4W)
2x - 2.2µF Electrolytic Capacitor (16V)
1x - 10µF Electrolytic Capacitor (16V)
1x - 10µF Electrolytic Capacitor (16V)
1x - 10mH Inductor
1x - 8 Ohm Speaker (1/4W)
1x - 9V Voltage Battery

View All Circuits

[TOP]

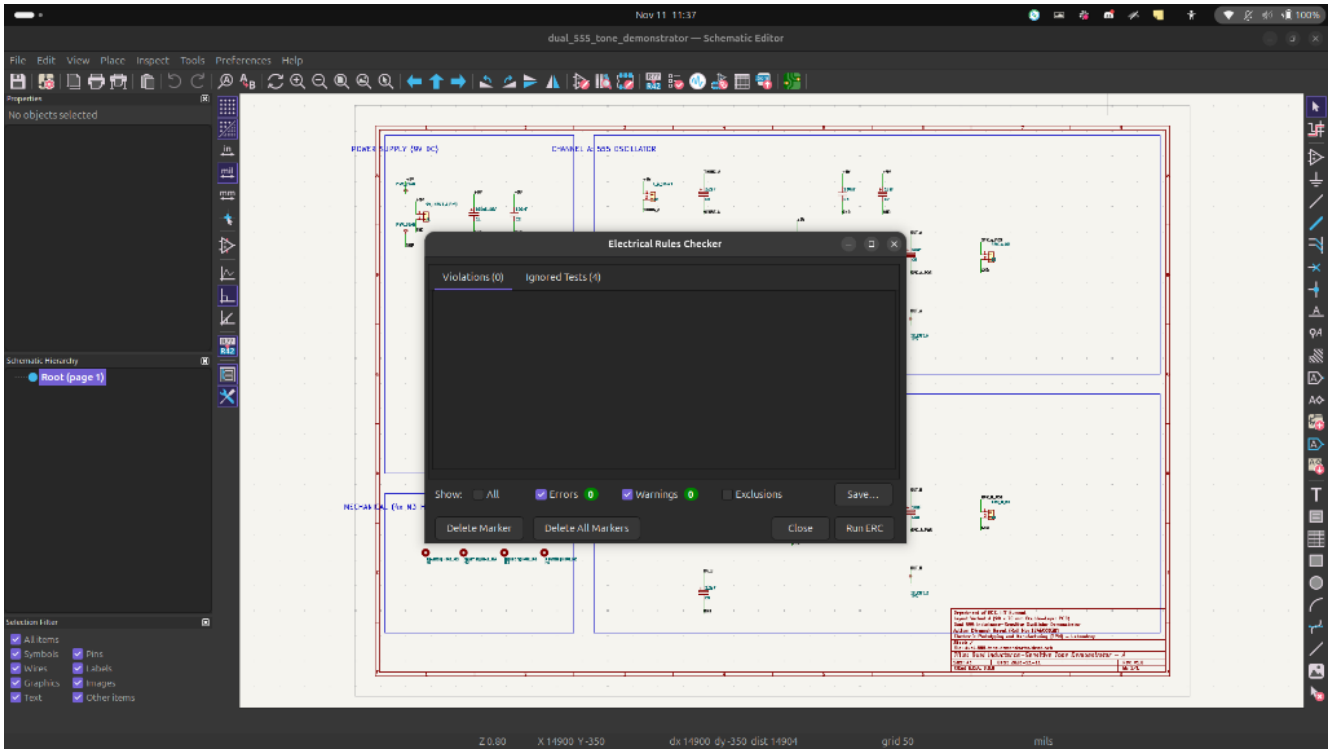
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Enlarged Circuit Schematic:



8. SCREENSHOT OF ELECTRICAL RULES CHECK (ERC)

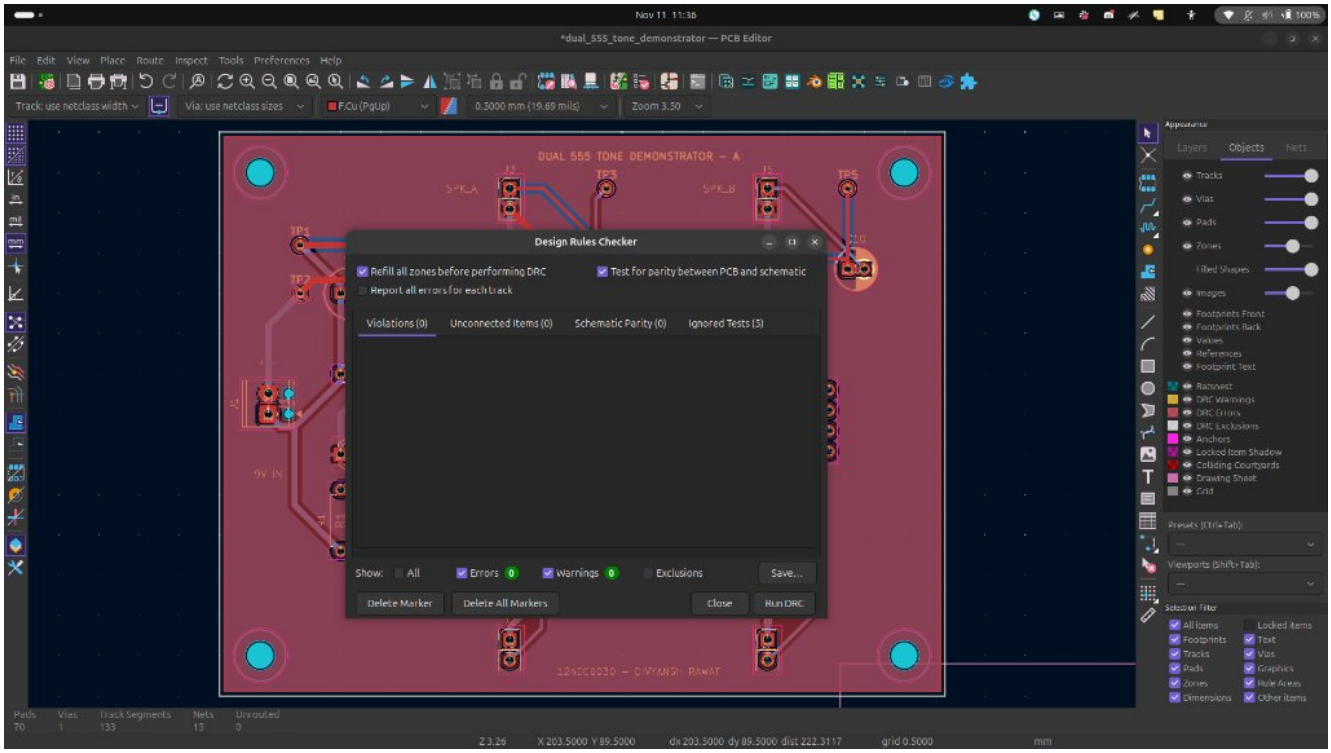
Live KiCad Schematic Editor Electrical Rules Checker dialog showing **0 Errors, 0 Warnings, and 0 Violations**.



ERC Status:	**PASSED (0 Errors, 0 Warnings)**	**KiCad Version:**	9.0.8+dfsg-1 (Linux x86_64)
Schematic File:	dual_555_tone_demonstrator.kicad_sch	**Sheet Name:**	Root Sheet (/)

9. SCREENSHOT OF DESIGN RULES CHECK (DRC)

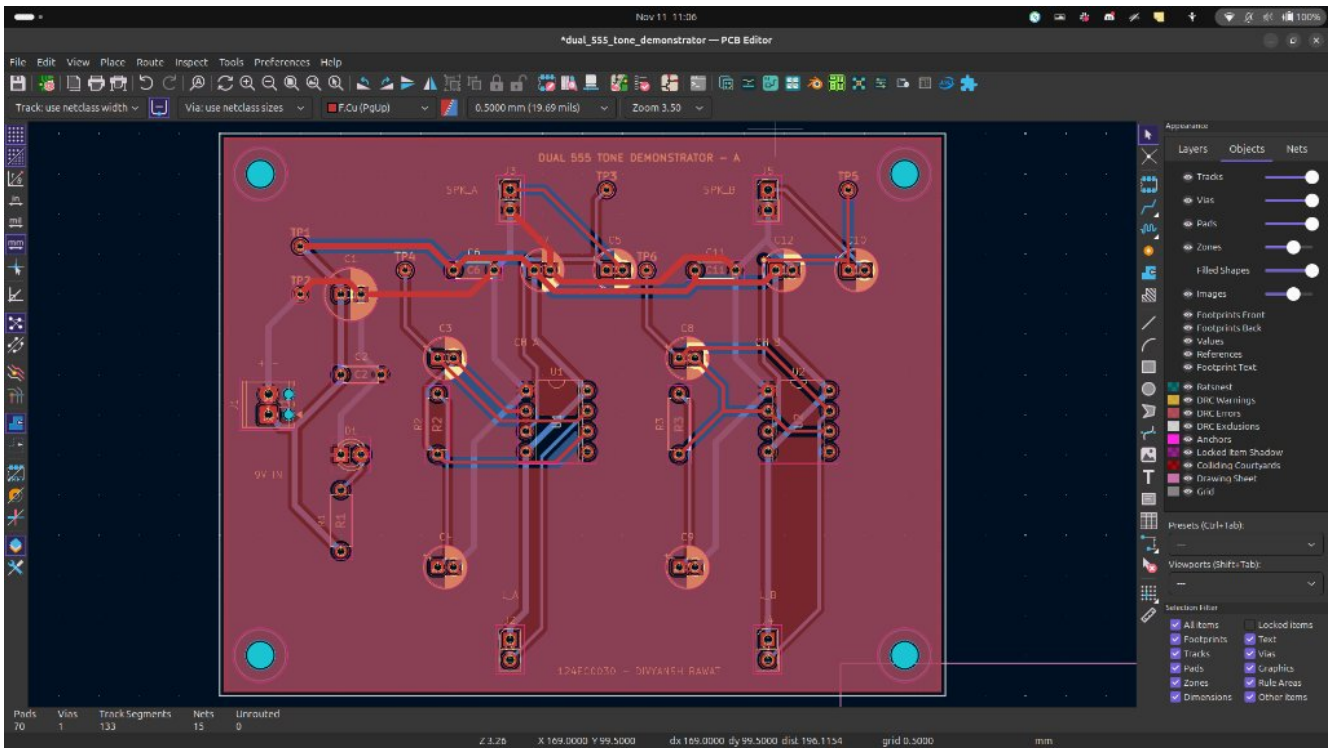
Live KiCad PCB Editor Design Rules Checker dialog showing **0 Violations, 0 Unconnected Items, and 0 Schematic Parity Warnings**.



DRC Violations:	**0 Violations**	**Unconnected Items:**	**0 Unconnected Items**
Schematic Parity:	**0 Issues (100% Match)**	**Overall DRC:**	**0 Errors, 0 Warnings**

3. LIVE COMPOSITE PCB LAYOUT (KiCad PCB Editor)

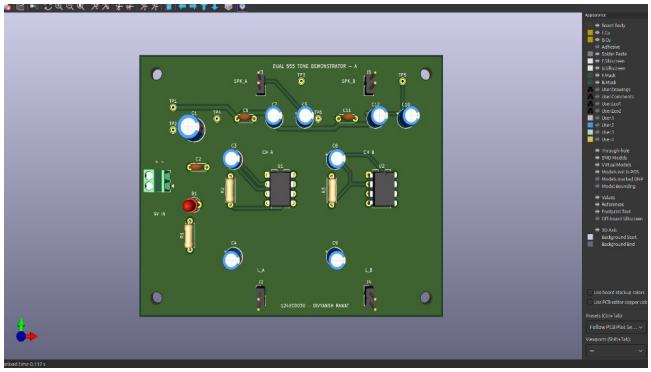
Live interactive PCB Editor screenshot showing double-sided routing (F.Cu red, B.Cu blue), continuous solid ground planes with thermal relief spokes, component placements, and Edge.Cuts outline.



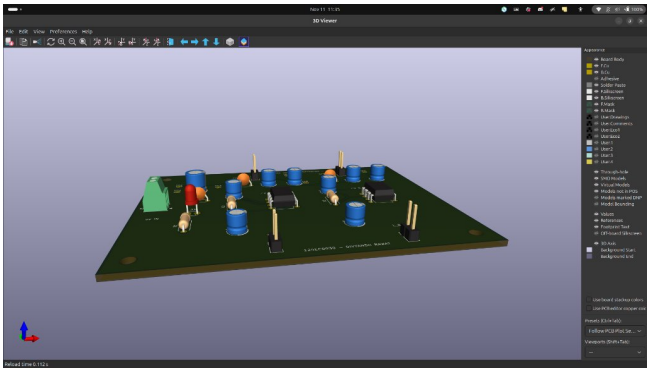
4. 3D VISUALIZATION VIEWS (REALISTIC GREEN SOLDERMASK)

Comprehensive 3D renders from multiple vantage points demonstrating component package fit, through-hole solder joints, connector orientations, and ground planes.

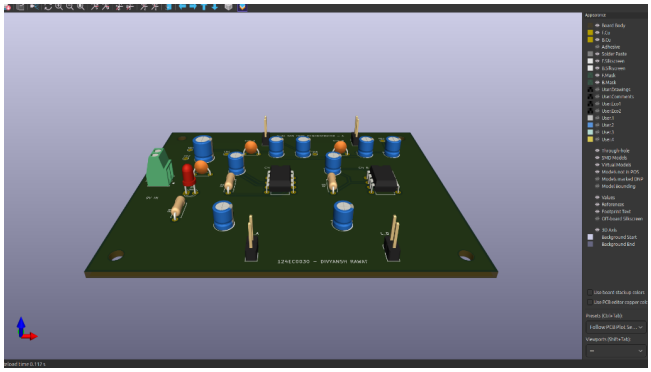
Top Orthogonal View:



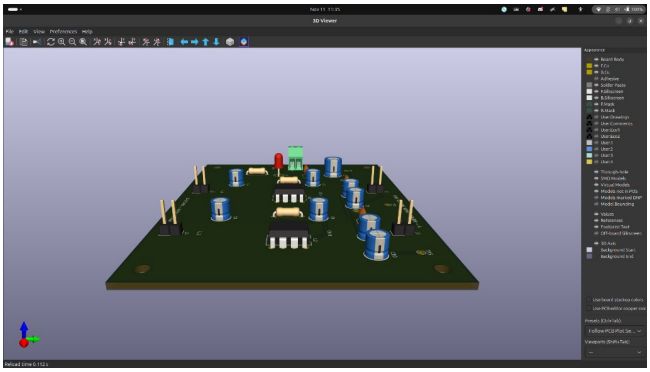
Isometric Perspective View:



Front Elevation View:



Side Profile View:



BILL OF MATERIALS (BOM)

Item	Reference	Value	Footprint	Qty	Description / Function
1	U1, U2	NE555P	DIP-8_W7.62mm	2	Precision Bipolar Timer IC
2	J1	9V_INPUT	TerminalBlock_MPT-0.5-2_P2.54mm	1	DC Power Input Terminal Block
3	J2, J4	L_A, L_B	PinHeader_1x02_P2.54mm_Vertical	2	External Inductor (10 mH) Connectors
4	J3, J5	SPK_A, SPK_B	PinHeader_1x02_P2.54mm_Vertical	2	External Speaker (8 Ohm) Connectors
5	C1	100uF_16V	CP_Radial_D6.3mm_P2.50mm	1	Power Supply Bulk Reservoir Capacitor
6	C2, C6, C11	100nF	C_Disc_D4.3mm_W1.9mm_P5.00mm	3	High-Frequency Ceramic Bypass Capacitors
7	C3, C4, C8, C9	2.2uF_16V	CP_Radial_D5.0mm_P2.00mm	4	Astable Timing Electrolytic Capacitors
8	C5, C7, C10, C12	10uF_16V	CP_Radial_D5.0mm_P2.00mm	4	AC Audio Output Coupling Capacitors
9	R1	1k	R_Axial_DIN0207_P7.62mm	1	Power LED Current-Limiting Resistor
10	R2, R3	47k	R_Axial_DIN0207_P7.62mm	2	Astable Timing Feedback Resistors
11	D1	PWR_LED	LED_D3.0mm	1	Green Power Status Indicator
12	TP1-TP6	TestPoint	TestPoint_THTPad_D1.5mm	6	Diagnostic Test Points (+9V, GND, Outs, Timings)
13	H1-H4	M3 Hole	MountingHole_3.2mm_M3	4	M3 Mechanical Chassis Mounting Holes

POWER SUPPLY (9V DC)

+9V
PWR_FLAG
+9V
9V_INPUT_J_PWR
1
2
J1
PWR_FLAG
GND
GND
+9V
100uF_16V
C1
GND
+9V
100nF
C2
GND
+9V
PWR_LED_GREEN
GND
D1
NetLED_PWR
+9V
R1
1k
NetLED_PWR
+9V
TP_9V
TP1
GND
TP_GND
TP2

MECHANICAL (4x M3 HOLES)

MountingHole_M3
H1
MountingHole_M3
H2
MountingHole_M3
H3
MountingHole_M3
H4

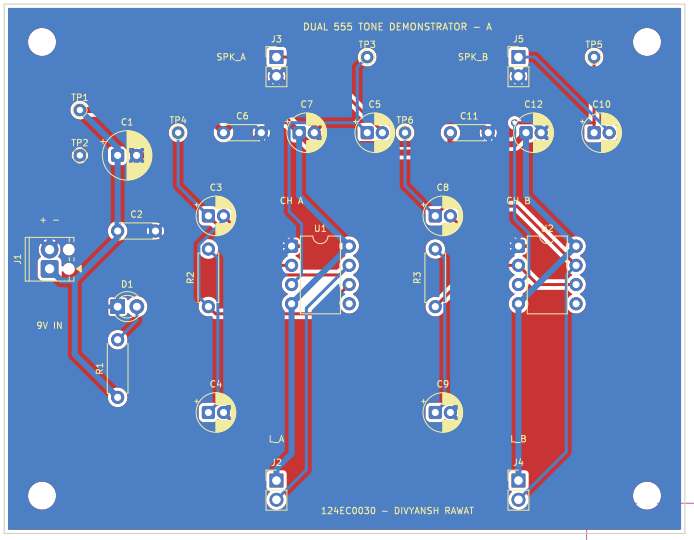
CHANNEL A: 555 OSCILLATOR

+9V
L_A_10mH
1
2
J2
TIMING_A
TIMING_A
2.2uF
C3
SENSE_A
+9V
100nF
C6
GND
+9V
10uF
C7
GND
TIMING_A
TP_TIM_A
TP4
SENSE_A
47k
R2
RC_A
+9V
5
8
+9V
4
RST
TIMING_A
DISCH
SENSE_A
SENSE_A
THRES
TRIG
CONT
VCC
OUT
3
OUT_A
GND
1
U1
NE555P
GND
OUT_A
10uF
C5
SPK_A_POS
SPK_A_POS
SPK_A_8R
1
2
J3
GND
RC_A
2.2uF
C4
GND
OUT_A
TP_OUT_A
TP3

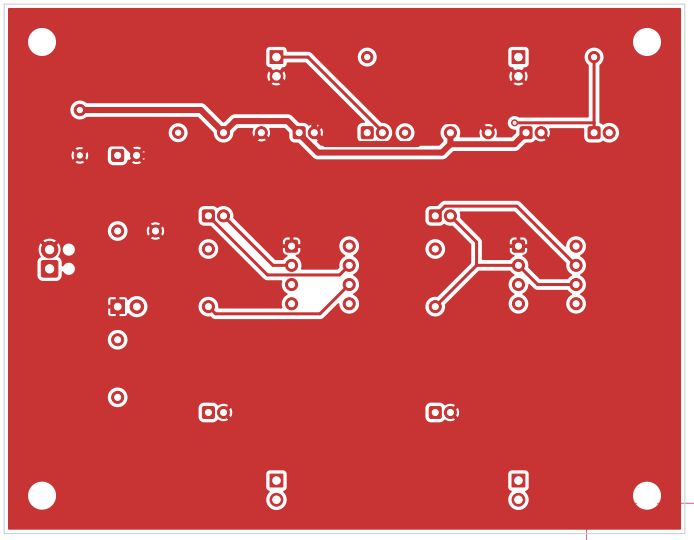
CHANNEL B: 555 OSCILLATOR

+9V
L_B_10mH
1
2
J4
TIMING_B
TIMING_B
2.2uF
C8
SENSE_B
+9V
100nF
C11
GND
+9V
10uF
C12
GND
TIMING_B
TP_TIM_B
TP6
SENSE_B
47k
R3
RC_B
+9V
5
8
+9V
4
RST
TIMING_B
DISCH
SENSE_B
SENSE_B
THRES
TRIG
CONT
VCC
OUT
3
OUT_B
GND
1
U2
NE555P
GND
OUT_B
10uF
C10
SPK_B_POS
SPK_B_POS
SPK_B_8R
1
2
J5
GND
RC_B
2.2uF
C9
GND
OUT_B
TP_OUT_B
TP5

Department of ECE, IIIT Kurnool
Layout Variant A (90 x 70 mm Double-Layer PCB)
Dual 555 Inductance-Sensitive Oscillator Demonstrator
Author: Divyansh Rawat (Roll No: 124EC0030)
Electronic Prototyping and Manufacturing (EPM) – Laboratory
Sheet: /
File: dual_555_tone_demonstrator.kicad_sch
Title: Dual Inductance-Sensitive Tone Demonstrator – A
Size: A3 | Date: 2026-11-11 | **Rev: v1.0**
KiCad E.D.A. 9.0.8 | Id: 1/1



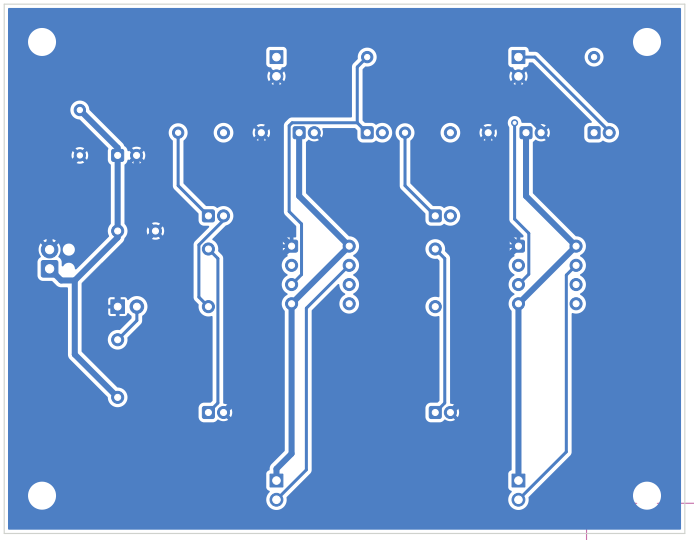
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.8		Id: 1/1



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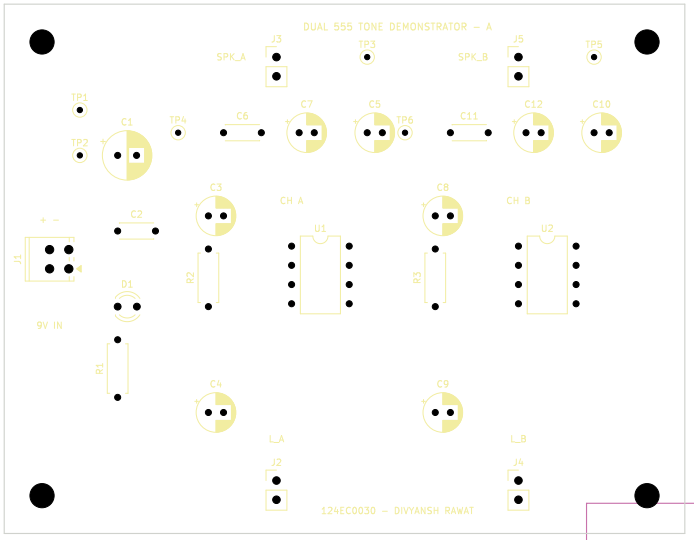
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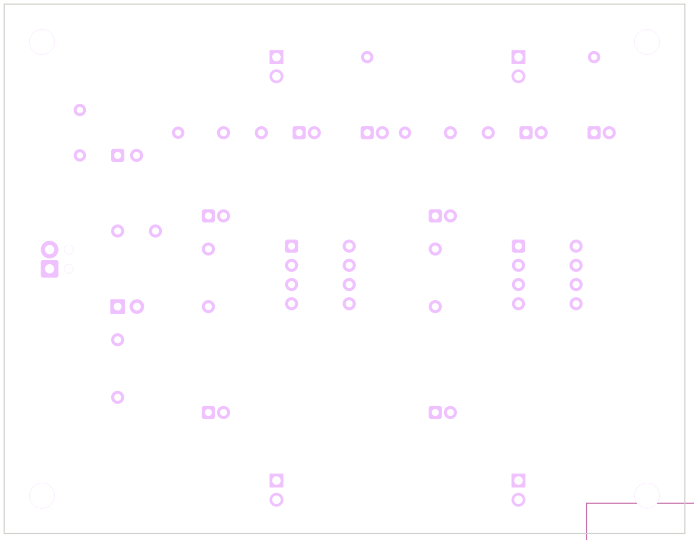
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Title:

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Sheet:		
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.8		Id: 1/1



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KiCad E.D.A. 9.0.8		Id: 1/1